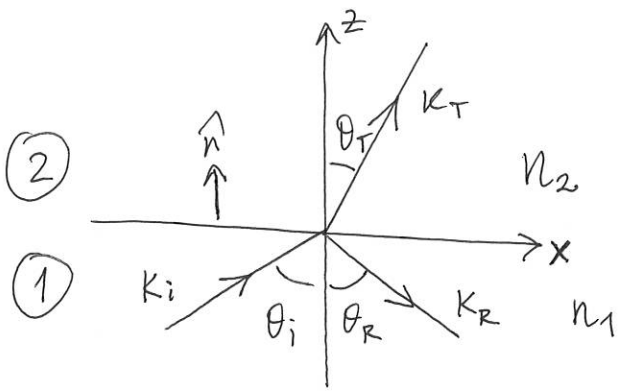



BC: $D_n(0+) - D_n(0-) = \sigma_f \to 0$ for $\sigma_f = 0$

$B_n(0+) - B_n(0-) = 0$

$E_{\parallel}(0+) - E_{\parallel}(0-) = 0$

$H_{\parallel}(0+) - H_{\parallel}(0-) = \hat{K}_f \times \hat{n} \to 0$ $\quad K_f = 0$

no delta-function like densities on the surface

$$\vec{E}_1(r,t) = \vec{E}_i\, e^{i\bar{k}_i \bar{r} - i\omega t} + \vec{E}_R\, e^{i\bar{k}_R \cdot \bar{r} - i\omega t}$$

$$\vec{E}_2(r,t) = \vec{E}_T\, e^{i\bar{k}_T \cdot \bar{r} - i\omega t}$$

BC require that on $z=0$ surface (interface between the media)

$$\vec{k}_i \cdot \vec{r}\,\big|_{z=0} = \vec{k}_R \cdot \vec{r}\,\big|_{z=0} = \vec{k}_T \cdot \vec{r}\,\big|_{z=0}$$

Choose $k_{iy} = k_{Ry} = k_{Ty} = 0$ (plane of incidence = xz plane)

$\Rightarrow$ (1) $k_{ix} = k_{Rx} = \frac{n_1 \omega}{c} \sin\theta_i = k_1 \sin\theta_i$

$\Rightarrow k_{Rx} = k_1 \sin\theta_R \Rightarrow \theta_R = \theta_i$

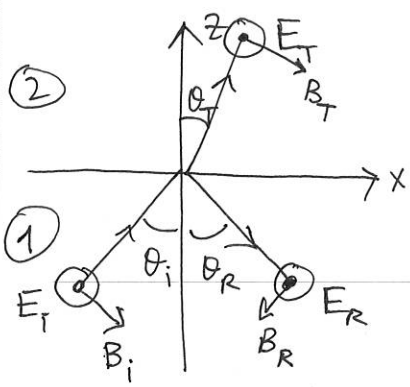
(2) $k_{ix} = k_{Tx} \Rightarrow k_1 \sin\theta_i = k_2 \sin\theta_T$

$$\frac{\omega}{c} n_1 \sin\theta_i = \frac{\omega}{c} n_2 \sin\theta_T$$

Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

S-polarization: $\vec{E} = (0, E_y, 0)$ , $\vec{B} = (B_x, 0, B_z)$

($\vec{E} \perp XZ$ plane)




$$E_y = e^{i k_{ix} x}\left(E_i e^{i k_{iz} z} + E_R e^{-i k_{iz} z}\right) \quad z<0$$

$$(k_{ix} = k_{Rx})$$

$$E_y = e^{i k_{Tx} x} E_T e^{i k_{Tz} z} \quad z>0$$

$$(k_{Tx} = k_{ix})$$

At $z=0$ $\quad E_{\parallel}(z=+0) = E_{\parallel}(z=-0) \Rightarrow E_i + E_R = E_T$

$$H_{\parallel}(z=+0) = H_{\parallel}(z=-0) \quad \left(H = \frac{B}{\mu}\right)$$

$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow$ ~~$\vec{k} \times \vec{E} = \omega B = k_x E_y \hat{z} - k_z E_y \hat{x}$~~

~~$\Rightarrow B_{1x} = -\frac{k_z}{\omega} E_y(z=0)$~~

$$i\omega \vec{B} = -\hat{x}\frac{\partial E_y}{\partial z} + \hat{z}\frac{\partial E_y}{\partial x}$$

$$B_{1x} = -\frac{1}{i\omega}\frac{\partial}{\partial z} E_y\Big|(z=-0) = \left(\frac{-i k_{iz}}{i\omega} E_i + \frac{i k_{iz}}{i\omega} E_R\right) e^{i k_{ix} x}$$

$$= -\frac{k_i \cos\theta_i}{\omega}\left(E_i - E_R\right) e^{i k_{ix} x}$$

$$B_{2x} = -\frac{1}{i\omega}\frac{\partial}{\partial z} E_y\Big|_{z=+0} = -\frac{k_{Tz}}{\omega} E_T e^{i k_{Tx} x}$$

$$H_{\parallel}(z=+0) = -\frac{k_T \cos\theta_T}{\mu_2 \omega} E_T e^{i k_{Tx} x}$$

$$H_{\parallel}(z=-0) = -\frac{k_i \cos\theta_i}{\mu_1 \omega}\left(E_i - E_R\right) e^{i k_{ix} x}$$

$\frac{k_i}{\omega} = \frac{n_1}{c}$ but $n_1 = \sqrt{\frac{\varepsilon_1 \mu_1}{\varepsilon_0 \mu_0}} = c\sqrt{\varepsilon_1 \mu_1} \Rightarrow \frac{n_1}{c} = \sqrt{\varepsilon_1 \mu_1}$

$$B_{2x} = -\frac{\partial_z E_{2y}}{i\omega} = -\frac{k_{Tz}}{\omega}\left(e^{ik_{Tx}x + ik_{Tz}z} E_T\right)$$

$$= -\cos\theta_T \frac{n_2}{c}\left( \qquad \right)$$

$\Rightarrow$ $H_{1x} = H_{2x}$ gives $-\cos\theta_i \frac{n_1}{\mu_1 c}\left(E_i - E_R\right) e^{ik_{ix}x}$

$$= -\cos\theta_T \frac{n_2}{\mu_2 c} E_T e^{ik_{Tx}x}$$

$n_1 = c\sqrt{\mu_1 \varepsilon_1} \Rightarrow \frac{n_1}{\mu_1 c} = \sqrt{\frac{\varepsilon_1}{\mu_1}} = Z_1^{-1}$ (intrinsic impedance)

$\hookrightarrow Z = \sqrt{\frac{\mu}{\varepsilon}}$

$$\cos\theta_i \frac{(E_i - E_R)}{Z_1} = \cos\theta_T \frac{E_T}{Z_2}$$

$$\begin{cases} Z_2 (E_i - E_R)\cos\theta_i = Z_1 E_T \cos\theta_T \\ E_i + E_R = E_T \end{cases}$$

$$(E_R + E_i) Z_1 \cos\theta_T = (E_i - E_R)\cos\theta_i Z_2$$

S polarization

$$E_R = \frac{\cos\theta_i Z_2 - \cos\theta_T Z_1}{\cos\theta_i Z_2 + \cos\theta_T Z_1} E_i$$

$$E_T = \frac{2\cos\theta_i Z_2}{\cos\theta_i Z_2 + \cos\theta_T Z_1} E_i$$

say, for normal incidence $\theta_i = \theta_T = 0$

$\Rightarrow E_R \sim \frac{Z_2 - Z_1}{Z_2 + Z_1} E_i$

$= \frac{n_1 - n_2}{n_1 + n_2} E_i$

$\Downarrow$

changes sign (phase $\pi$) on reflecting from more dense medium!

From $n_1 \sin\theta_i = n_2 \sin\theta_T$

find $\cos\theta_T = \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2} = \frac{1}{n_2}\sqrt{n_2^2 - n_1^2 \sin^2\theta_i}$

<u>Non-magnetic</u> media: $\mu_1 = \mu_2 = \mu_0 \Rightarrow Z_j = \sqrt{\frac{\mu_j}{\varepsilon_j}} = \sqrt{\frac{\mu_0}{\varepsilon_j}} =$

$$= \frac{\mu_0}{\sqrt{\varepsilon_j \mu_0}} = \frac{\mu_0 c}{n_j} \quad (j = 1, 2)$$

$$E_R = \frac{\frac{\cos\theta_i}{n_2} - \frac{\cos\theta_T}{n_1}}{\frac{\cos\theta_i}{n_2} + \frac{\cos\theta_T}{n_1}} E_i = \frac{n_1\cos\theta_i - n_2\cos\theta_T}{n_1\cos\theta_i + n_2\cos\theta_T} E_i$$

$$E_T = \frac{2 n_1 \cos\theta_i}{n_1\cos\theta_i + n_2\cos\theta_T} E_i$$

Energy transport — intensities of reflected / refracted waves

$$\langle \vec{S} \rangle = \frac{1}{2} \mathrm{Re}\left(\vec{E} \times \vec{H}^*\right) = \frac{1}{2Z} |\vec{E}|^2 \hat{k}$$

$\Rightarrow$ Reflection $R_s = \frac{|E_R|^2}{|E_i|^2} = \left(\frac{n_1\cos\theta_i - n_2\cos\theta_2}{n_1\cos\theta_i + n_2\cos\theta_T}\right)^2$

(normal component of Poynting vector)

Transmission $T_s = \frac{\langle S_{z,2}\rangle}{\langle S_{z,1}\rangle} = \frac{\frac{\cos\theta_T}{Z_2}|E_T|^2}{\frac{\cos\theta_i}{Z_1}|E_i|^2} =$

$$= \frac{Z_1}{Z_2}\frac{\cos\theta_T}{\cos\theta_i}\left|\frac{E_T}{E_i}\right|^2 \longrightarrow \frac{n_2\cos\theta_2}{n_1\cos\theta_1}\left|\frac{E_T}{E_i}\right|^2$$

$$= \frac{Z_1\cos\theta_T}{Z_2\cos\theta_i}\frac{\cos^2\theta_i \cdot 4Z_2^2}{(\cos\theta_i Z_2 + \cos\theta_T Z_1)^2} = \frac{4 Z_1 Z_2 \cos\theta_i \cos\theta_T}{(Z_2\cos\theta_i + Z_1\cos\theta_T)^2}$$

Energy flux $\vec{S} = \frac{1}{\mu} \vec{E} \times \vec{B} = \vec{E} \times \vec{H}$

$$S_z = \frac{1}{\mu}\left(E_x B_y - E_y B_x\right) = -\frac{1}{\mu} E_y B_x \quad \text{for s-polarization}$$

$$S_{1z} = -\frac{1}{\mu_1} \frac{1}{2}\left(E_{1y} + E_{1y}^*\right) \frac{1}{2}\left(B_{1x} + B_{1x}^*\right) = -\frac{1}{2\mu_1} E_{1y} B_{1x}^*$$

time-average

$$S_{iz} = -\frac{1}{2\mu_1} E_{iy} B_{ix}^* = -\frac{1}{2\mu_1} E_i \left(-\frac{k_{iz}}{\omega} E_i^*\right) = \frac{k_i \cos\theta_i}{2\mu_1 \omega} |E_i|^2$$

$$S_{Rz} = -\frac{1}{2\mu_1} E_{Ry} B_{Rx}^* = -\frac{1}{2\mu_1} E_R \left(\frac{k_{iz}}{\omega} E_R^*\right) = -\frac{k_i \cos\theta_i}{2\mu_1 \omega} |E_R|^2$$

$$S_{Tz} = -\frac{1}{2\mu_2} E_{Ty} B_{Tx}^* = -\frac{1}{2\mu_2} E_T \left(-\frac{k_{Tz}}{\omega} E_T^*\right) = \frac{k_T \cos\theta_T}{2\mu_2 \omega} |E_T|^2$$

Observe that $S_{Rz}$ moves away from the interface ($S_{Rz} < 0$).

Reflection coefficient (reflectance)

$$R_s = \left|\frac{S_{Rz}}{S_{iz}}\right| = \frac{|E_R|^2}{|E_i|^2} = \left|\frac{n_1 \cos\theta_i - n_2 \cos\theta_T}{n_1 \cos\theta_i + n_2 \cos\theta_T}\right|^2 \quad \text{for } \mu_1 = \mu_2 = \mu_0 \text{ medium}$$

s-polarization

Transmission coefficient (transmittance)

$$T_s = \left|\frac{S_{Tz}}{S_{iz}}\right| = \frac{k_T \cos\theta_T \, \mu_1}{k_i \cos\theta_i \, \mu_2} \frac{|E_T|^2}{|E_i|^2}$$

$$k_T = \frac{\omega}{v_2} = \frac{\omega}{c} n_2$$

$$k_i = \frac{\omega}{v_1} = \frac{\omega}{c} n_1$$

$$T_s = \frac{n_2 \mu_1 \cos\theta_T}{n_1 \mu_2 \cos\theta_i} \frac{|E_T|^2}{|E_i|^2} = \sqrt{\frac{\varepsilon_2}{\mu_2}} \sqrt{\frac{\mu_1}{\varepsilon_1}} \frac{\cos\theta_T}{\cos\theta_i} \frac{|E_T|^2}{|E_i|^2} =$$

$$= \frac{Z_1}{Z_2} \frac{\cos\theta_T}{\cos\theta_i} \left( \frac{2\cos\theta_i Z_2}{\cos\theta_i Z_2 + \cos\theta_T Z_1} \right)^2 = \frac{4 Z_1 Z_2 \cos\theta_i \cos\theta_T}{(Z_2 \cos\theta_i + Z_1 \cos\theta_T)^2}$$

$\Rightarrow$ for non-magnetic media with $\mu_1 = \mu_2 = \mu_0$

$$\Rightarrow \quad Z = \sqrt{\frac{\mu_0}{\varepsilon}} = \frac{\mu_0}{\sqrt{\varepsilon \mu_0}} = \frac{\mu_0 c}{n}$$

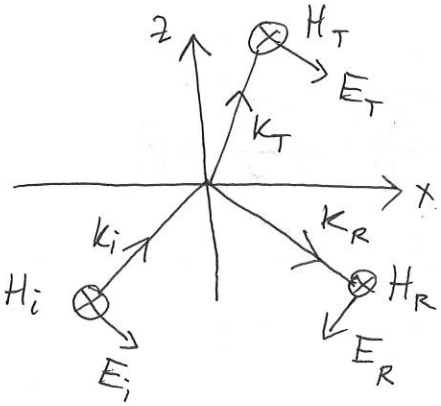
$$\Rightarrow \quad T_s = \frac{4 n_1 n_2 \cos\theta_i \cos\theta_T}{(n_1 \cos\theta_i + n_2 \cos\theta_T)^2}$$

Therefore $R_s + T_s = 1$.

For $\mu_j = \mu_0$, $T_s = \dfrac{4 n_1 n_2 \cos\theta_i \cos\theta_T}{(n_1 \cos\theta_i + n_2 \cos\theta_T)^2}$

Observe $T_s + R_s = 1$.

<u>P-polarization</u>




($\vec{H} \perp XZ$ plane)

$\vec{H} = (0, H_y, 0)$

$H_y = e^{ik_{ix}x}\left(H_i e^{ik_{iz}z} + H_R e^{-ik_{iz}z}\right)_{z=0}$

$H_y = e^{ik_{Tx}x} H_T e^{ik_{Tz}z}$

$\Rightarrow H_i + H_R = H_T$ (1), $E_{ix} + E_{Rx} = E_{Tx}$ (2)

$E_i \cos\theta_i - E_R \cos\theta_i = E_T \cos\theta_T$

Use $\vec{\nabla} \times \vec{H} = \dfrac{\partial \vec{D}}{\partial t} = \varepsilon \dfrac{\partial \vec{E}}{\partial t}$ $\left(\Rightarrow -i\omega\varepsilon \vec{E} = \hat{x}(-\partial_z H_y) + \hat{z}(\partial_x H_y)\right)$

$-i\omega\varepsilon_1 E_{ix} = -\partial_z H_i = -i k_{iz} H_i$ <u>at $z=0$</u>

$E_{ix} = \dfrac{k_{iz}}{\varepsilon_1 \omega} H_i$, $E_{Rx} = -\dfrac{k_{iz}}{\varepsilon_1 \omega} H_R = -\cos\theta_i Z_1 H_R$

$E_{Tx} = \dfrac{k_{Tz}}{\varepsilon_2 \omega} H_T = \cos\theta_T Z_2 H_T$

$\dfrac{k}{\omega} = \dfrac{k}{\frac{c}{n}k} = \dfrac{n}{c} = \sqrt{\varepsilon\mu}$

$\Rightarrow Z_1 \cos\theta_i (H_i - H_R) = Z_2 \cos\theta_T H_T$ (2')

$Z_2 \cos\theta_T (H_i + H_R) = Z_1 \cos\theta_i (H_i - H_R)$

$\Rightarrow H_R = \dfrac{Z_1 \cos\theta_i - Z_2 \cos\theta_T}{Z_1 \cos\theta_i + Z_2 \cos\theta_T} H_i$, $H_T = \dfrac{2 Z_1 \cos\theta_i}{Z_1 \cos\theta_i + Z_2 \cos\theta_T} H_i$

$$\langle \vec{S} \rangle = \frac{1}{2} \mathrm{Re}\left\{ \vec{E} \times \vec{H}^* \right\} = \frac{1}{2Z} |\vec{E}|^2 \hat{k} = \frac{Z}{2} |\vec{H}|^2 \hat{k}$$

b/c $-i\,\omega\varepsilon \vec{E} = \hat{x}\,(-i k_z H_y) + \hat{z}\,(i k_x H_y)$

$k_z = k\cos\theta \qquad k_x = k\sin\theta$

$$\vec{E} = \hat{x}\,\frac{k}{\omega\varepsilon}\cos\theta\, H_y + \hat{z}\left(-\frac{k}{\omega\varepsilon}\sin\theta\, H_y\right) = (\cos\theta\,\hat{x} - \sin\theta\,\hat{z})\, Z\, H_y$$

$$\Rightarrow |\vec{E}| = Z\, H_y$$

Therefore $R_p = \left|\frac{H_R}{H_i}\right|^2 = \left(\frac{Z_1\cos\theta_i - Z_2\cos\theta_T}{Z_1\cos\theta_i + Z_2\cos\theta_T}\right)^2.$

$$T_p = \frac{\frac{1}{2} Z_2 |H_T|^2 \cos\theta_T}{\frac{1}{2} Z_1 |H_i|^2 \cos\theta_i} = \frac{Z_2\cos\theta_T}{Z_1\cos\theta_i}\left(\frac{2 Z_1\cos\theta_i}{Z_1\cos\theta_i + Z_2\cos\theta_T}\right)^2$$

$$= \frac{4 Z_1 Z_2 \cos\theta_i \cos\theta_T}{(Z_1\cos\theta_i + Z_2\cos\theta_T)^2} \longrightarrow \frac{4 n_1 n_2 \cos\theta_i \cos\theta_T}{(n_2\cos\theta_i + n_1\cos\theta_T)^2}$$

(for $\mu_i = \mu_0$)

<u>Normal incidence</u> $\theta_i = 0 \Rightarrow \theta_T = 0$:

$$R_s = R_p = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2, \quad T_s = T_p = \frac{4 n_1 n_2}{(n_1 + n_2)^2}.$$

* Absence of reflection (at Brewster's angle) for P-polarization:

$$H_R = 0 \Rightarrow Z_1\cos\theta_i = Z_2\cos\theta_2 \Rightarrow n_2\cos\theta_i = n_1\cos\theta_T$$

$$\left(\frac{n_2}{n_1}\right)^2 \cos^2\theta_i = \cos^2\theta_T = 1 - \left(\frac{n_1}{n_2}\right)^2 \sin^2\theta_i \qquad 1 = \cos^2 + \sin^2$$

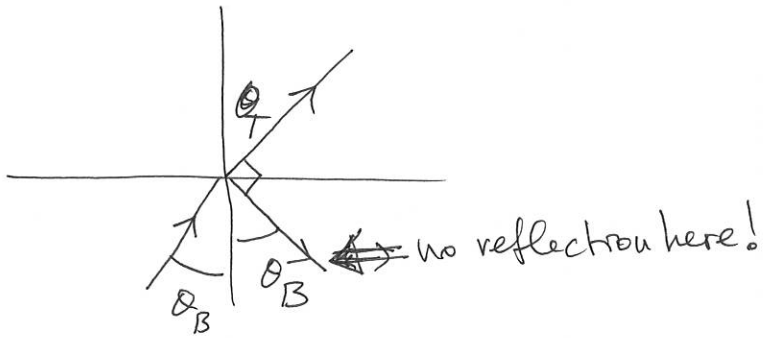
$$\left(\frac{n_2^2}{n_1^2} - 1\right)\cos^2\theta_i = \left(1 - \frac{n_1^2}{n_2^2}\right)\sin^2\theta_i \Rightarrow \frac{\sin^2\theta_i}{\cos^2\theta_i} = \frac{n_2^2}{n_1^2}$$

$$\Rightarrow \theta_i = \theta_B : \tan\theta_B = \frac{n_2}{n_1}$$

$$\frac{\sin\theta_B}{\cos\theta_B} = \frac{n_2}{n_1} \quad \text{but} \quad \frac{n_1}{n_2} = \frac{\sin\theta_T}{\sin\theta_B}$$

$$\Rightarrow \frac{\sin\theta_B}{\cos\theta_B} \cdot \frac{\sin\theta_T}{\sin\theta_B} = 1 \Rightarrow \sin\theta_T = \cos\theta_B$$

$$\Rightarrow \theta_B = \frac{\pi}{2} - \theta_T$$




For $\theta_i = \theta_B$ and <u>p-polarization</u> $H_R = 0 \Rightarrow E_R = 0$

$\Rightarrow$ (1) no reflection.

(2) s-polarized light is reflected

$\Rightarrow$ "polarization by reflection"

$\Rightarrow$ at $\theta_B$ only s-polarization is reflected.

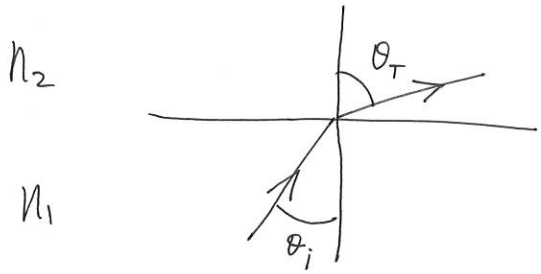
$\Downarrow$

reflected light is polarized parallel to the interface ($\vec{E} \parallel$ XY plane)

(+ total internal reflection)

Total internal reflection $(n_1 > n_2)$

$$n_1 \sin\theta_i = n_2 \sin\theta_T$$




But $\sin\theta_T = \frac{n_1}{n_2}\sin\theta_i \to 1$ for $\sin\theta_{ic} = \frac{n_2}{n_1} < 1$:

at this $\theta_c = \theta_{ic}$ the refracted wave propagates at $\theta_T = \pi/2$

i.e. along the interface boundary.

For $\theta_i > \theta_c \longrightarrow$ total internal reflection.

Snell's law: $k_{2x} = k_{1x} = k_1 \sin\theta_i = \frac{\omega n_1}{c}\sin\theta_i$

$$\Rightarrow \quad k_{2z} = \sqrt{k_2^2 - k_{2x}^2} = \sqrt{\frac{\omega^2 n_2^2}{c^2} - \frac{\omega^2 n_1^2}{c^2}\sin^2\theta_i} = \frac{\omega n_2}{c}\sqrt{1 - \left(\frac{n_1 \sin\theta_i}{n_2}\right)^2}$$

$$= \frac{\omega n_2}{c}\sqrt{1 - \left(\frac{\sin\theta_i}{\sin\theta_c}\right)^2} = i\varkappa$$

$$\varkappa = \frac{\omega n_2}{c}\sqrt{\left(\frac{\sin\theta_i}{\sin\theta_c}\right)^2 - 1} > 0 \quad \text{for } \theta_i > \theta_c.$$

P-polariz. $\vec{E}_2(x,z) = \vec{E}_T(x,z) = \vec{E}_T\, e^{ik_{2x}x + ik_{2z}z - i\omega t}$

$$= \vec{E}_T\, e^{ik_{2x}x - i\omega t}\, e^{-\varkappa z}$$ decays exponentially into medium 2.

What are $\vec{E}_T$ components? $\vec{\nabla}\cdot\vec{E}_2(x,z) = 0$ $(\varepsilon_2 = \text{const})$

$$\Rightarrow \quad E_{Tx}\, ik_{2x} - E_{Tz}\,\varkappa = 0$$

$$\Rightarrow \quad \frac{E_{Tz}}{E_{Tx}} = \frac{ik_{2x}}{\varkappa} = \frac{ik_1 \sin\theta_1}{\varkappa} = i\,\frac{n_1 \sin\theta_i \sin\theta_c}{n_2\sqrt{\sin^2\theta_i - \sin^2\theta_c}}$$

Physical field $\vec{E}_2 = Re\left\{ \vec{E}_T(x,z) \right\} =$

$$= Re\left\{ \left( E_{Tx}\,\hat{x} + \frac{i k_1 \sin\theta_1}{\varkappa} E_{Tx}\,\hat{z} \right) e^{-\varkappa z}\, e^{i(k_{2x}x - \omega t)} \right\}$$

$$= E_{Tx}\, e^{-\varkappa z} \cos(k_{2x}x - \omega t)\,\hat{x} - \frac{k_1 \sin\theta_1}{\varkappa} E_{Tx}\, e^{-\varkappa z} \sin(k_{2x}x - \omega t)\,\hat{z}.$$

let $\mu_j = \mu_0$: $\quad r_p \equiv \frac{H_R}{H_i} = \frac{Z_1 \cos\theta_i - Z_2 \cos\theta_T}{Z_1 \cos\theta_i + Z_2 \cos\theta_T} \rightarrow \frac{n_2 \cos\theta_i - n_1 \cos\theta_T}{n_2 \cos\theta_i + n_1 \cos\theta_T}$

$$\left.\begin{aligned} k_{2z} &= k_2 \cos\theta_T \\ \cos\theta_T &= \frac{i\varkappa}{k_2} \end{aligned}\right\} \Rightarrow r_p = \frac{n_2 k_2 \cos\theta_i - i n_1 \varkappa}{n_2 k_2 \cos\theta_i + i n_1 \varkappa} = e^{-i\delta}.$$

$\Rightarrow R_p = |r_p|^2 = 1 \Rightarrow$ this is full reflection indeed.

Simply: $\cos\theta_T = \sqrt{1 - \sin^2\theta_T} = \sqrt{1 - \left(\frac{k_1}{k_2}\sin\theta_i\right)^2} = \sqrt{1 - \left(\frac{n_1 \sin\theta_i}{n_2}\right)^2}$

$k_2 \sin\theta_T = k_1 \sin\theta_i \quad , \quad k_1 = \frac{\omega}{v_1} = \frac{\omega n_1}{c}$

$k_2 = \frac{\omega}{v_2} = \frac{\omega n_2}{c}$

$$\cos\theta_T = \sqrt{1 - \left(\frac{\sin\theta_i}{\sin\theta_c}\right)^2} = \frac{i\varkappa}{k_2}$$

Conductors $\frac{\varepsilon(\omega)}{\varepsilon_0} = \frac{\varepsilon_b(\omega)}{\varepsilon_0} + \frac{i\sigma(\omega)}{\varepsilon_0\omega} \approx 1 - \frac{\omega_p^2}{\omega(\omega+\frac{i}{\tau})} = 1 - \frac{\omega_p^2}{\omega^2(1+i\gamma)}$

$$\sigma(\omega) = \frac{\sigma_0}{1-i\omega\tau}, \quad \sigma_0 = \frac{e^2 n\tau}{m} \qquad \gamma \equiv \frac{1}{\omega\tau}$$

$$\sigma(\omega) = \frac{i\varepsilon_0\omega_p^2}{\omega+\frac{i}{\tau}}$$

For $\omega \ll \omega_p$, but arbitrary $\omega\tau$, $\frac{\varepsilon(\omega)}{\varepsilon_0} \approx -\frac{\omega_p^2}{\omega^2(1+i\gamma)}$

$$\Rightarrow n(\omega) = \frac{\omega_p}{\omega}\frac{1}{\sqrt{-1-i\gamma}} \qquad \left(\text{neglecting } \frac{\varepsilon_b}{\varepsilon_0} \sim 1\right)$$

Need to choose solution which decays inside the conductor $\Rightarrow$ $n''(\omega) > 0$

$$\frac{1}{\sqrt{-1-i\gamma}} = \sqrt{\frac{-1+i\gamma}{(-1-i\gamma)(-1+i\gamma)}} = \sqrt{\frac{-1+i\gamma}{1+\gamma^2}} = \frac{e^{i\varphi/2}}{(1+\gamma^2)^{1/4}}$$

$$-1+i\gamma = (1+\gamma^2)^{1/2}e^{i\varphi} \Rightarrow \begin{cases}(1+\gamma^2)^{1/2}\cos\varphi = -1\\ (1+\gamma^2)^{1/2}\sin\varphi = \gamma\end{cases}$$

$$\tan\varphi = -\gamma$$

$$\varphi = -\tan^{-1}(\gamma) + \pi k$$

$$n''(\omega) \sim \mathrm{Im}\left(e^{+i\varphi/2}\right) = \mathrm{Im}\left(e^{-\frac{i}{2}\tan^{-1}(\gamma)+\frac{i}{2}\pi k}\right)$$

Suppose $\gamma \ll 1 \Rightarrow \tan^{-1}(\gamma) \approx \gamma$

$$\mathrm{Im}\left(e^{-\frac{i}{2}\gamma}\right) = -\sin\frac{\gamma}{2} = -\frac{\gamma}{2} < 0$$

but $\mathrm{Im}\left(e^{i\frac{\pi}{2}-i\frac{\gamma}{2}}\right) = \sin\left(\frac{\pi}{2}-\frac{\gamma}{2}\right) = \cos\frac{\gamma}{2} > 0$

Therefore we choose $\varphi = \pi - \tan^{-1}(\gamma)$

Thus $n(\omega) = \frac{\omega_p}{\omega} \frac{e^{i\frac{\pi}{2} - \frac{i}{2}\tan^{-1}(\gamma)}}{(1+\gamma^2)^{1/4}} = n'(\omega) + i\, n''(\omega)$

$$n''(\omega) = \frac{\omega_p}{\omega} \frac{\sin\left[\frac{\pi}{2} - \frac{1}{2}\tan^{-1}(\gamma)\right]}{(1+\gamma^2)^{1/4}} = \frac{\omega_p \cos\left(\frac{1}{2}\tan^{-1}(\gamma)\right)}{(1+\gamma^2)^{1/4}\omega}$$

$$n'(\omega) = \frac{\omega_p}{\omega(1+\gamma^2)^{1/4}} \cos\left(\frac{\pi}{2} - \frac{1}{2}\tan^{-1}(\gamma)\right) = \frac{\omega_p \sin\left(\frac{1}{2}\tan^{-1}(\gamma)\right)}{(1+\gamma^2)^{1/4}\omega}$$

<u>Limits:</u> (a) $\gamma = \frac{1}{\omega\tau} \ll 1$ (so that $\omega\tau \gg 1$) $\Rightarrow \varepsilon''$ very small.
but $\omega \ll \omega_p$

$\Rightarrow \tan^{-1}(\gamma) \to \gamma \ll 1$

$$n''(\omega) \sim \frac{\omega_p}{\omega} \gg 1$$

$$n'(\omega) \sim \frac{\omega_p}{\omega} \sin\left(\frac{\gamma}{2}\right) \doteq \left(\frac{1}{2}\gamma\right)\left(\frac{\omega_p}{\omega}\right) \ll n''(\omega)$$

(b) $\gamma = \frac{1}{\omega\tau} \gg 1 \Rightarrow \omega\tau \ll 1 \Rightarrow \arctan(\gamma) \to \frac{\pi}{2}$

$$\cos\left[\frac{1}{2}\arctan(\gamma)\right] = \frac{1}{\sqrt{2}}$$

$$\sin\left[\frac{1}{2}\arctan(\gamma)\right] = \frac{1}{\sqrt{2}}$$

$$\Rightarrow n(\omega) = \frac{\omega_p}{\sqrt{\gamma}\,\omega} e^{i\frac{\pi}{2} - i\frac{\pi}{4}} = \frac{\omega_p e^{i\pi/4}}{\sqrt{\gamma}\,\omega}$$

$\Rightarrow$ here $n'(\omega) = n''(\omega) = \frac{\omega_p}{\sqrt{2\gamma}\,\omega} \gg 1$

Since $n_2 \gg 1$ we have $n_1 \sin\theta_i = n_2 \sin\theta_T$

$\Rightarrow \sin\theta_T = \frac{n_1 \sin\theta_i}{n_2} \ll 1 \Rightarrow$ refracted wave propagates almost along the normal, $\theta_T \ll 1$.

Consider <u>normal incidence</u> $\theta_i = \theta_T = 0$:

$$R_s = R_p = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 = \left|\frac{n_2' + i n_2'' - n_1}{n_2' + i n_2'' + n_1}\right|^2 =$$

$$= \frac{(n_2' - n_1)^2 + (n_2'')^2}{(n_2' + n_1)^2 + (n_2'')^2} \approx \frac{(n_2')^2\left(1 - \frac{n_1}{n_2'}\right)^2 + (n_2'')^2}{(n_2')^2\left(1 + \frac{n_1}{n_2'}\right)^2 + (n_2'')^2} =$$

$n_2' = n_2''$
$\gamma \gg 1$

$$= \frac{1 + \left(1 - \frac{n_1}{n_2'}\right)^2}{1 + \left(1 + \frac{n_1}{n_2'}\right)^2} = \frac{1 + 1 - 2n_1/n_2'}{2 + 2n_1/n_2'} = \frac{1 - n_1/n_2'}{1 + n_1/n_2'}$$

$$= 1 - \frac{2n_1}{n_2'} = 1 - 2n_1 \frac{\omega\sqrt{2\gamma}}{\omega_p} = 1 - \frac{n_1}{\omega_p}\sqrt{\frac{8\omega}{\tau}}$$

$$\omega_p = \sqrt{\frac{e^2 n}{m \varepsilon_0}}, \quad \sigma_0 = \frac{e^2 n \tau}{m} = \varepsilon_0 \omega_p^2 \tau$$

$$\Rightarrow 1 - R = n_1 \sqrt{\frac{8\omega}{\omega_p^2 \tau}} = n_1 \sqrt{\frac{8\varepsilon_0 \omega}{\sigma_0}} = \sqrt{\frac{8\varepsilon_1 \omega}{\sigma_0}}$$

Using $n_1 = \sqrt{\frac{\varepsilon_1}{\varepsilon_0}}$

Hagen – Rubens relation: $R(\omega) \approx 1 - \sqrt{\frac{8\varepsilon_1 \omega}{\sigma_0}}$

high reflectivity
(makes metals shiny)

Q: We know that for $\omega < \omega_p$ the EM wave does <u>not</u> propagate inside the conductor $\Rightarrow$ why $1 - R \neq 0$?

A: It must be energy lost due to Joule's heating! (homework problem)

$$1 - R = 1 - \frac{(n_2' - n_1)^2 + (n_2'')^2}{(n_2' + n_1)^2 + (n_2'')^2} = \frac{4 n_1 n_2'}{(n_2' + n_1)^2 + (n_2'')^2} \approx \frac{4 n_1 n_2'}{(n_2')^2 + (n_2'')^2} =$$

$$= 2 n_1 / n_2' \; !!!$$